TRACK 1 · AI/ML — OPEN INNOVATION · STUDENT TRACK

TrapKiller

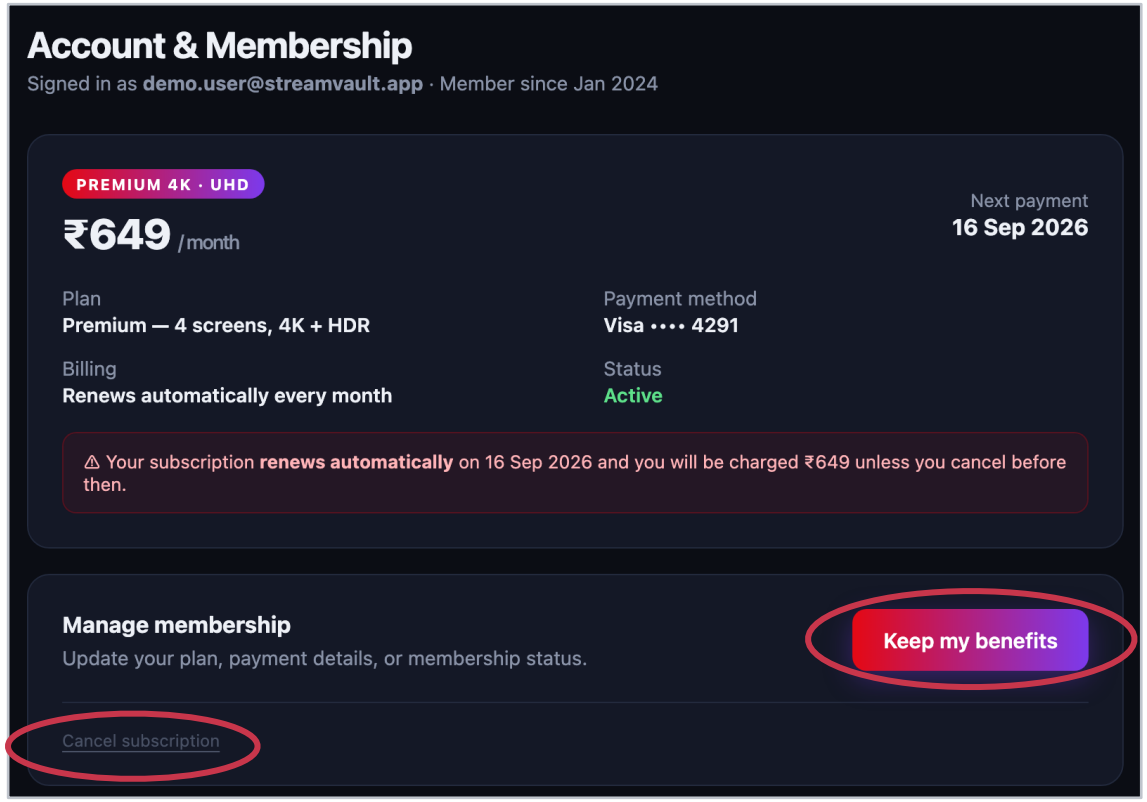
An AI agent that spots dark patterns on subscription pages, walks the cancel maze for you, and asks before the last click.

Detects 13 dark-pattern types, each backed by a quote from the page

Acts Steps around retention traps to reach the real cancel screen

Scales with TinyFish Hosted web agents run sweeps across thousands of sites

Team Panda · Track 1: AI/ML — Open Innovation



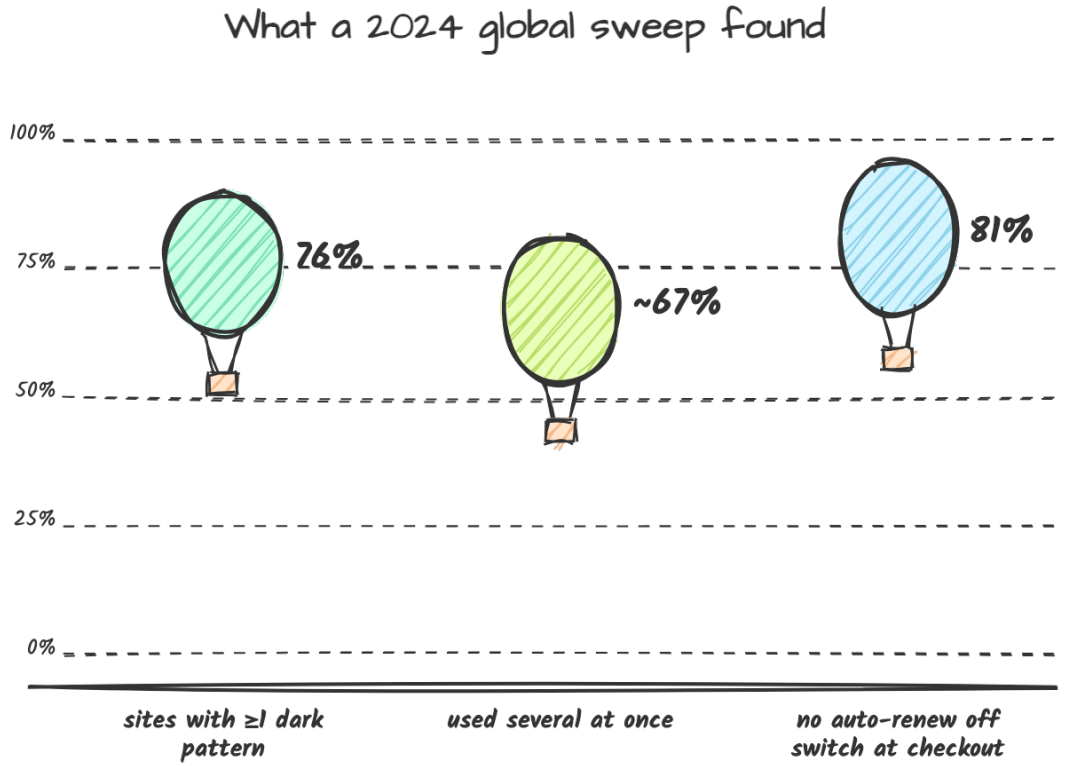
Roach motel: the exit is a faint link

Misdirection: the loud button keeps you paying

Our StreamVault test site, built with the same tricks real subscription pages use

01 · PROBLEM STATEMENT

Most subscription sites are built to keep you paying



Why it's still unsolved

1. **India has rules, not reach.**
   13 dark patterns are banned since Nov 2023, but checking a site is still manual, one page at a time.

2. **Every maze is fought alone.**
   Cancel links are hidden, the decline button guilt-trips, and auto-renew is on by default.

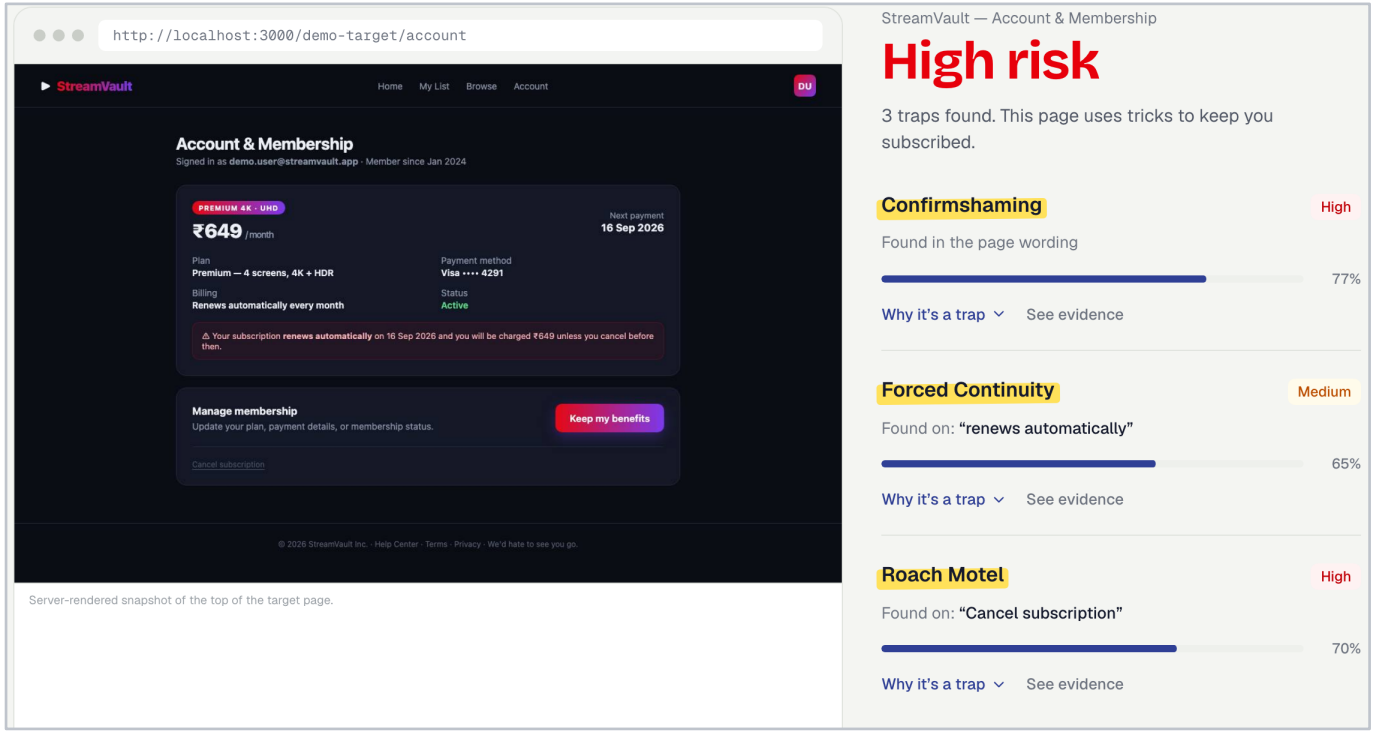
3. **Tools either see or act, never both.**
   Blockers ignore cancel flows. Generic AI browser agents click straight through traps without flagging them.

Sources: FTC / ICPEN / GPEN review of 642 subscription sites and apps, July 2024 · CCPA Guidelines for Prevention and Regulation of Dark Patterns, 30 Nov 2023

02 · PROPOSED SOLUTION

Scan the trap, walk the cancel flow, stop for your approval

Real output, not a mockup



Scanner run, 20 Oct 2026, on our test site: High risk, 3 traps, each tied to page text.

One product, three moves

1 Scan

Read-only verdict. Every finding shows its evidence quote and a confidence score.

2 Cancel

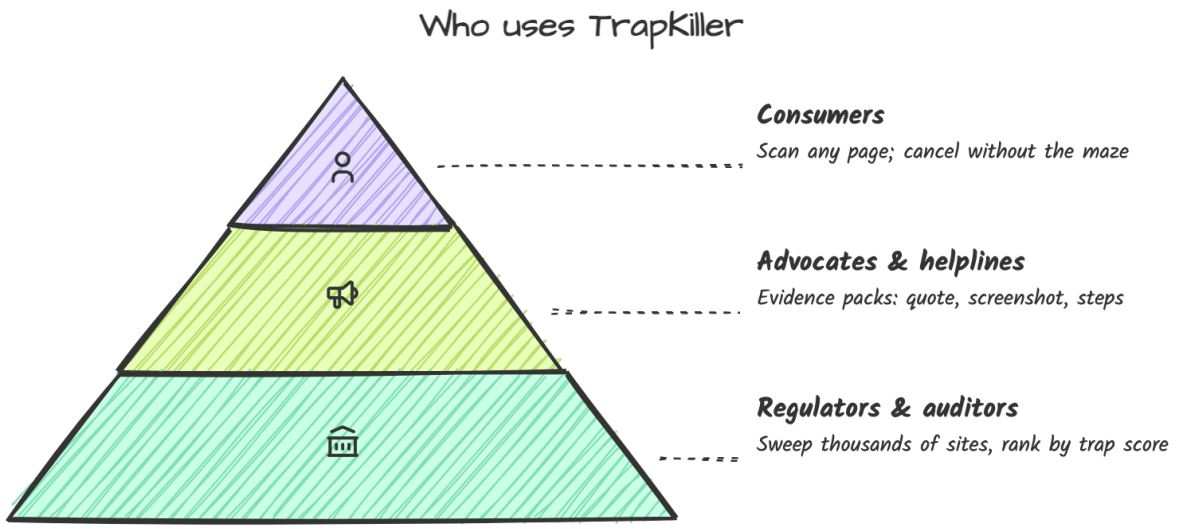
The agent reads each screen, declines retention offers, and heads for the real exit.

3 Prove

It parks the final irreversible click until you approve, then saves a proof screenshot.

03 · TARGET USERS AND USE CASE

Built for the person stuck in the maze, useful to those who police it



Use case: one stubborn subscription

1. **Pastes the account page URL** of a streaming service she no longer watches.
2. **TrapKiller flags 3 traps** a guilt-trip decline, a buried cancel link, silent auto-renew.
3. **The agent walks the flow** declines two retention offers, stops on the final confirm screen.
4. **She taps Approve** proof screenshot and every step are saved, ready if she is charged again.

Illustrative scenario; the flow matches what the agent does on the test site today.

04 · ROLE OF MACHINE LEARNING AND AI

AI decides what is a trap; rules and people keep it honest

Four jobs AI does

Neural detector

MiniLM embeddings + softmax head; 94.0% on 463 held-out texts

Grounded LLM auditor

a finding is kept only if its quote is on the page

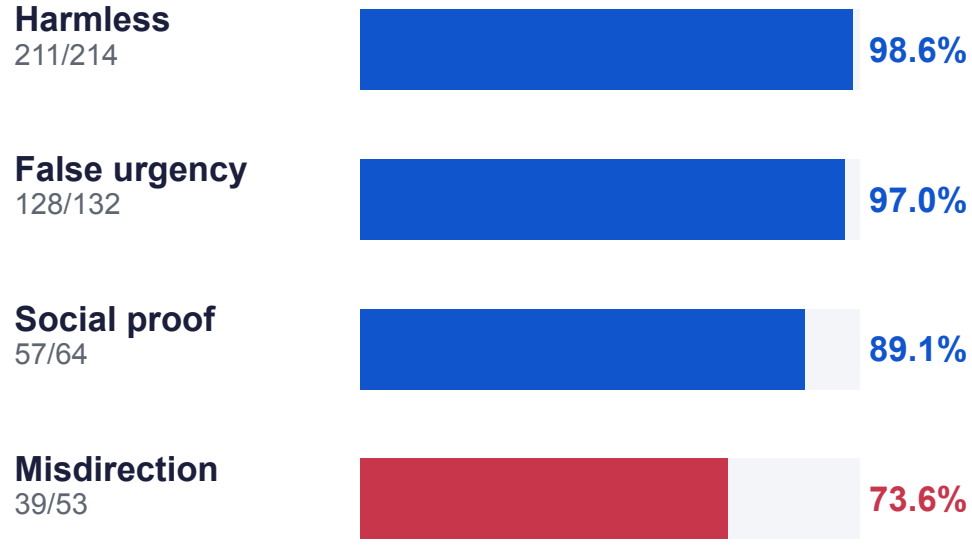
Route learning

verified clicks are rewarded; planner follows the shortest known path

Human feedback

'Is this really a dark pattern?' votes retrain the signal

Held-out accuracy by class (n = 463)

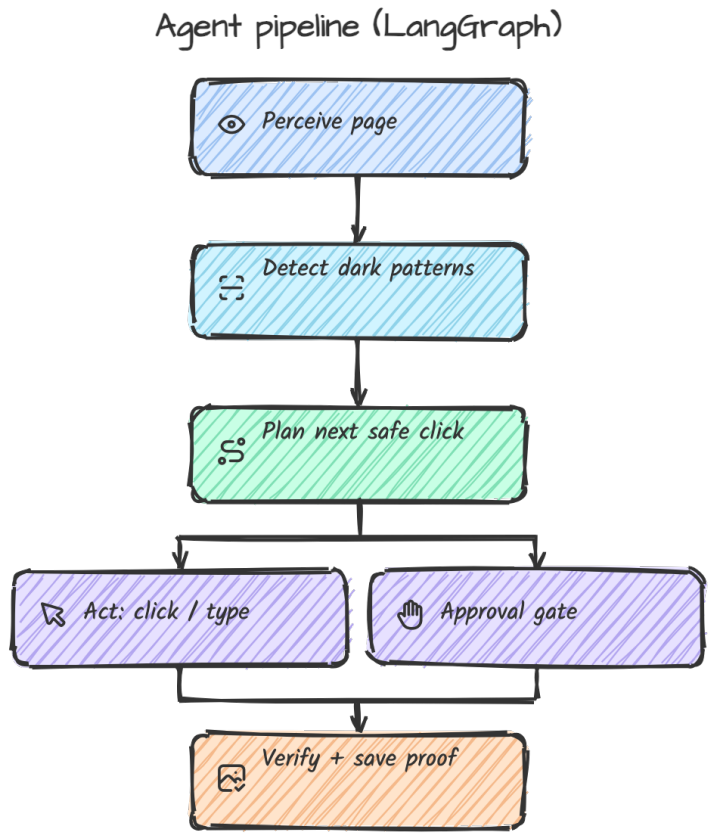


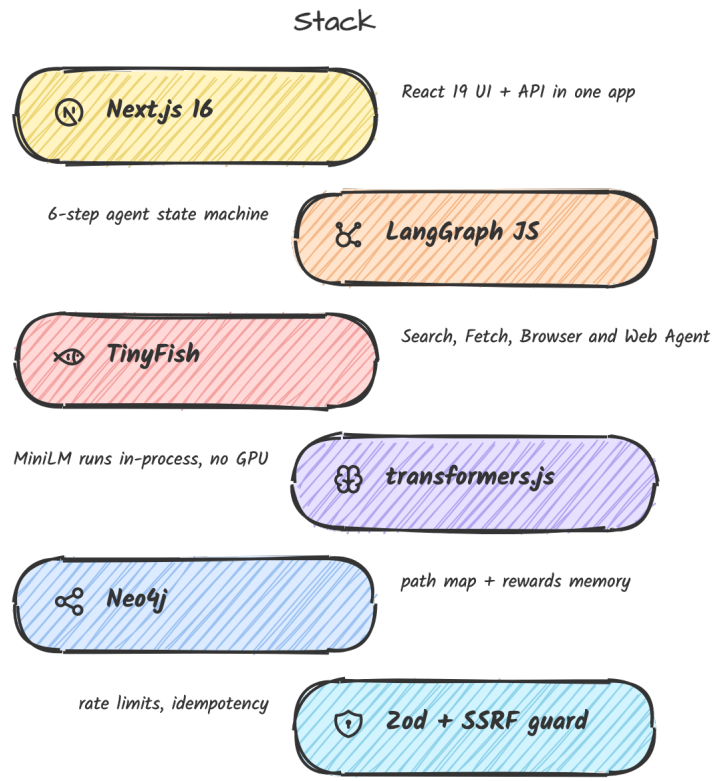
Misdirection is our weakest class. So the model only reports findings above 85% confidence, and phrase rules plus a grounded LLM back it up.

Neural head trained on the ec-darkpattern dataset (Yada et al., IEEE BigData 2022): 1,850 training texts, 463 held out, 94.0% overall accuracy

05 · PROPOSED TECHNICAL APPROACH

A six-step agent loop on a stack two students can run on a laptop





Safety rule: the agent can only click, type, scroll and wait. Page text is data, never instructions.

06 · TINYFISH INTEGRATION

All four TinyFish capabilities, each doing one real job

1 Search API
Finds the way in

User types only "StreamVault". Search returns the account, cancel and help pages, plus public complaints about the service.

2 Fetch API
Feeds the detector

Fetches those pages as clean text (JS rendered, 10 URLs per call). Our neural detector scans every sentence for traps.

3 Browser
Runs our own agent

Our LangGraph agent drives a TinyFish cloud browser through Playwright over CDP, instead of a local Chromium.

4 Web Agent
Walks flows at scale

Natural-language goal: "decline offers, stop before final confirm". Runs live for one user, or in batches for a sweep.

ONE SCAN, END TO END



Blue = TinyFish. Dark = TrapKiller's own AI and safety step. TinyFish gives us the live web; we decide what is a trap and when to stop.

06 · TINYFISH INTEGRATION

How TrapKiller calls each TinyFish API

API	Call	What we send	What we read back	Plugs into TrapKiller
1 Search	**GET** api.search.tinyfish.ai	query="StreamVault cancel" purpose, language="en"	results[].url results[].title, snippet → cancel + help pages	api/scanner/analyze name → page URLs
2 Fetch	**POST** api.fetch.tinyfish.ai	urls=[up to 10 pages] format="markdown" links=true, ttl=0	results[].text results[].final_url errors[] per URL	detection/classifier.ts text → neural detector
3 Browser	**POST** api.browser.tinyfish.ai DELETE /{session_id}	url=<cancel page> timeout_seconds=300	session_id, cdp_url → connectOverCDP(cdp_url)	browser/manager.ts replaces local Chromium
4 Web Agent	**POST** agent.tinyfish.ai /v1/automation/run-sse	url, goal="Decline offers; STOP before confirm" max_steps=40	STREAMING_URL (live view) PROGRESS, COMPLETE /run-async for sweeps	agent/runner.ts queue/runner.ts (sweeps)

One scan: Search finds pages → Fetch gives clean text → our detector flags traps → Browser or Web Agent walks the flow → you approve → proof. If TinyFish is unreachable, the agent falls back to local Playwright.

Endpoints and fields from the TinyFish API reference (docs.tinyfish.ai). Every call sends X-API-Key: $TINYFISH_API_KEY from .env.local

07 · EXPECTED IMPACT

Each scan turns a hidden trick into evidence someone can act on

Who gains what

Consumers

Stop unwanted charges before the renewal date, without hours of clicking

Advocates & helplines

Complaint-ready packs: quote, screenshot, every step

Regulators

A ranked list of sites to inspect, instead of manual spot checks

The market

Public trap scores reward companies that design fairly

How TrapKiller compares

	Manual	Blocker extensions	Generic AI browser agent	TrapKiller
Flags dark patterns with evidence	—	—	○	●
Walks the cancel flow	●	—	●	●
Stops before the irreversible click	●	—	○	●
Saves a proof screenshot	○	—	○	●
Learns the shortest route over runs	—	—	—	●
Sweeps many sites at once	—	—	○	○*

● does it ○ partly — no * planned with TinyFish · Comparison is our own assessment of typical tools in each category

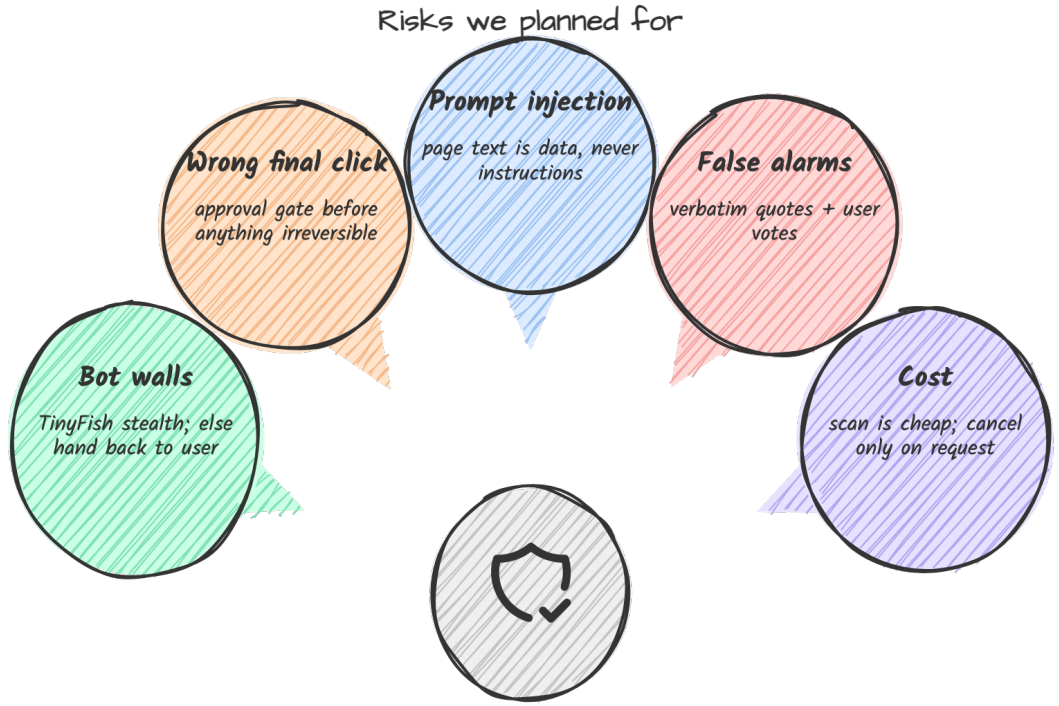
08 · FEASIBILITY

It already runs end to end; the risks have named fixes

Working today

- ✓ Full loop on the test site: scan → agent → approval pause → proof
- ✓ 74 of 74 automated tests pass
- ✓ Neural detector: 94.0% on held-out data, runs on CPU
- ✓ Zero paid services needed: LLM and Neo4j are optional
- ✓ 13-type taxonomy uses the names in India's 2023 guidelines

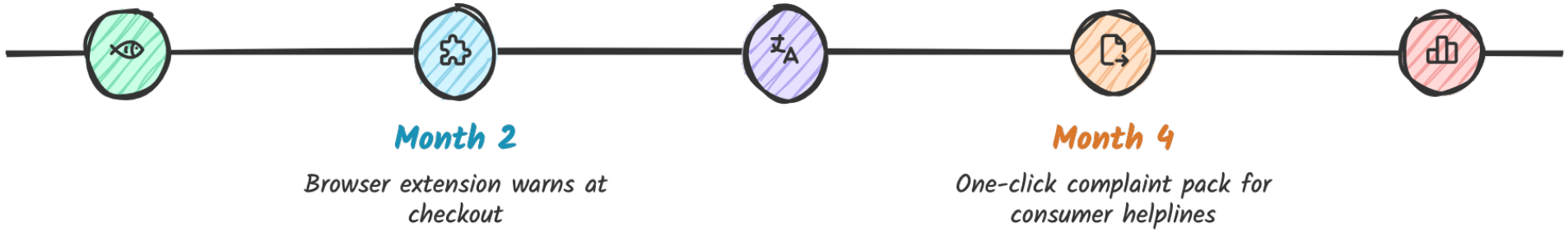
Being fixed now: an unverified run can still be reported as cancelled, and Reject/Stop are not fully wired to the running agent.



Checked on 20 Oct 2026: npm test 74/74 passing; scan and agent run against the StreamVault test site

09 · FUTURE SCOPE

From one test site to a public trap index



Our ask: TinyFish credits to run the first 500-site sweep, so we can publish India's first trap-score ranking of subscription services.